Nourin Shahin

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Summary

Applied AI Engineer specializing in large language models, agentic AI, and generative systems, with published IEEE/arXiv research and end-to-end experience taking LLM applications from prototype through fine-tuning, retrieval, rigorous evaluation, and production deployment. Designed and shipped a multi-agent, RAG-grounded HPC assistant powered by a domain-fine-tuned (QLoRA) model, served behind a REST API and deployed on GPU infrastructure. Built LLM benchmarking frameworks on Google Vertex AI and developed adversarial testing and prompt-optimization techniques published in IEEE. Disciplined about model evaluation—reproducibility, metrics, and failure-mode analysis—with a security mindset from benchmarking LLMs against the OWASP Top 10 for LLM Applications. Google Cloud and Azure AI certified; strong Python and scalable data-pipeline skills.

Technical Skills

Agentic AI & LLM Systems: Multi-Agent Orchestration, Tool-Calling / Agent Workflows, Retrieval-Augmented Generation (RAG), LLM Fine-Tuning (QLoRA / LoRA / PEFT), Instruction Tuning, Teacher–Student Distillation, Prompt Engineering & Optimization, LangChain, Stateful Conversational Memory
LLM Tooling & Frameworks: PyTorch, Hugging Face Transformers, PEFT, TRL, bitsandbytes, sentence-transformers, ChromaDB (vector search / embeddings), FastAPI, NLP, Deep Learning
Model Evaluation & MLOps Principles: Benchmarking Frameworks, BERTScore / ROUGE-L / Cosine / Domain-Specific Metrics, Performance Monitoring, Cross-Seed Reproducibility, Failure-Mode Analysis, Experiment Design, Pre-Production Model Testing
Cloud & Compute Infrastructure: Google Cloud (Vertex AI, IAM, Logging), Azure AI, AWS, SLURM Clusters, GPU / CUDA (A100), Distributed Computing (FABRIC Testbed, MPI)
AI Security & Privacy: OWASP Top 10 for LLM Applications, Adversarial & Encoding-Based Prompt Attacks, Threat Modeling, Data-Leakage Testing
Programming & Data: Python, SQL, Pandas, NumPy, Matplotlib, Bash, Scalable Data Pipelines; C++, Go, Rust, Java, Swift

Education

Texas A&M University–San Antonio — San Antonio, TX
Master of Science in Computer Science (GPA: 4.0) — *Jan 2025 – Expected Aug 2026*

Alexandria University — Alexandria, Egypt
Bachelor in Computer and Communication Engineering — *Graduated July 2024*

Experience

Research Assistant – LLM Systems, Agentic AI & Evaluation — Apr 2025 – Apr 2026
Texas A&M University–San Antonio — *San Antonio, TX*

- Designed and shipped an end-to-end **agentic AI assistant for HPC user support**: a multi-agent system in which an orchestrator coordinates dedicated retrieval, generation, and documentation-crawling agents, adds stateful multi-turn memory, and serves answers through a FastAPI backend and web interface—deployed on Jetstream2 A100 GPUs and exposed to end users over a secure tunnel.
- **Fine-tuned a domain-specialized LLM** with QLoRA (4-bit NF4) on a Qwen2.5-Coder-7B base, training on 4,000+ instruction pairs generated through **teacher–student distillation**: a 14B teacher model synthesized Q&A from documentation automatically crawled across 80+ university and national supercomputing centers.
- Engineered the **retrieval-augmented generation pipeline**—ChromaDB vector store with BGE-large embeddings, top-*k* semantic retrieval, and citation-grounded answers—with live knowledge-base ingestion (automated crawl and manual upload) so new clusters can be added at inference time without retraining.
- Built an end-to-end **LLM/SLM evaluation framework on Google Vertex AI**, benchmarking code-generation performance across 243 tasks and measuring execution time, memory usage, and cross-seed reproducibility—the pre-production model testing and performance-monitoring discipline required before any model reaches production.

- Evaluated LLM systems against **all OWASP Top 10 for LLM Applications** vulnerabilities, including training-data leakage and insecure output handling, and developed adversarial and encoding-based prompt attacks to probe and harden model behavior; documented failure modes and mitigations, published in IEEE and arXiv.
- Built scalable Python/SQL data pipelines and visualizations (Pandas, Matplotlib) to prepare experimental data and track model-performance metrics across configurations, supporting faculty publications and grant deliverables; contributed to peer code and experiment-design reviews.

HPC Research Intern – ML for Systems Jun 2026 – Present
Research Computing, Texas A&M University–San Antonio *San Antonio, TX*

- Researching **ML-driven optimization of computing workloads**—intelligent resource allocation, scheduling, and job management—framed as an applied optimization problem over heterogeneous GPU/CPU cluster resources.
- Developing and benchmarking ML models for **workload prediction**, measuring improvements in throughput, scalability, and efficiency; authoring a survey of AI-for-HPC methods across 50+ sources and presenting findings to the research team.

Graduate Assistant Nov 2025 – Present
College of Graduate Studies, Texas A&M University–San Antonio *San Antonio, TX*

- Provide institutional support and cross-department stakeholder communication while maintaining full research and coursework commitments.

iOS Mobile Application Developer Intern Jun 2023 – Aug 2023
Pharos Solutions *Alexandria, Egypt*

- Built and shipped educational iOS applications in Swift within a production codebase, participating in the full software development lifecycle—feature development, debugging, testing, and release.

Publications

N. Shahin, I. Alsmadi. “HPC-LLM: Practical Domain Adaptation and Retrieval-Augmented Generation for HPC Support.” *arXiv preprint*, 2026.

N. Shahin, I. Alsmadi. “Llama Model Security: From OWASP Top 10 Benchmarking to Encoding-Based Attacks.” *arXiv preprint*, 2026.

N. Shahin, I. Alsmadi. “Benchmarking LLAMA Model Security Against OWASP Top 10 For LLM Applications.” *IEEE*, 2026.

Certifications

Google Cloud Certified: Associate Cloud Engineer
Microsoft Certified: Azure AI Cloud Developer Associate
Additional: iOS Mobile Development (Pharos Solutions), Web Development (AMIT)

Awards & Grants

NAIRR Travel Grant: NAIRR Pilot Conference 2026
FABRIC Travel Grant: KNIT 12 Conference 2026
BRIDGE Travel Grant: PEARC26 Conference 2026
CAHSI Travel Grant: GMIS Conference 2025